Comparative Analysis of Machine Learning Algorithms for Breast Cancer Diagnosis Using the Wisconsin Breast Cancer Dataset

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This paper aims to apply and compare classification algorithms — Logistic Regression, Decision Tree, Random Forest, and KNN — to predict breast cancer diagnosis as malignant or benign, and identify the most influential features in early detection.

1. Introduction:

Breast cancer remains one of the most significant public health challenges of the twenty-first century, representing a leading cause of cancer-related morbidity and mortality among women worldwide. According to the International Agency for Research on Cancer (IARC), approximately 2.3 million new cases of breast cancer were recorded globally in 2022, accounting for nearly 24% of all cancer cases and 15% of all cancer-related deaths in women (IARC, 2024). Projections published in *Nature Medicine* (Kim et al., 2025) indicate that global breast cancer incidence is expected to rise by 38%, with related deaths increasing by as much as 68% by the year 2050 if current trends persist. These figures underscore the urgent need for scalable, accurate, and cost-effective approaches to early detection and diagnosis.

The prognosis of breast cancer is strongly dependent on the stage at which it is identified. When diagnosed at a localized stage, the five-year relative survival rate is approximately 99%; however, this figure falls sharply to 30% in cases of distant-stage disease (American Cancer Society, 2024). This stark disparity highlights the critical importance of early and accurate differentiation between malignant and benign tumors. Conventional diagnostic procedures, including mammography and fine needle aspiration (FNA) biopsy, are widely utilized in clinical settings; however, they are subject to inter-observer variability, resource constraints, and the inherent limitations of manual interpretation. Consequently, there is a growing body of research exploring the application of computational and data-driven methods to augment clinical decision-making in oncology.

In recent years, machine learning (ML) has emerged as a powerful paradigm in medical diagnostics, offering the ability to identify complex, non-linear patterns within high-dimensional clinical datasets that may elude conventional statistical analysis. Supervised classification algorithms, in particular, have demonstrated considerable promise in supporting the binary classification of tumors as either malignant or benign, based on quantitative morphological features extracted from tissue images. The integration of such algorithms into diagnostic workflows has the potential to improve classification accuracy, reduce diagnostic delays, and facilitate more equitable access to quality diagnostic support across diverse healthcare settings.

This study employs the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, originally compiled by Dr. William H. Wolberg at the University of Wisconsin–Madison and widely recognized as a benchmark dataset in biomedical machine learning research. The dataset comprises 569 samples, each described by 30 numerical features derived from digitized images of fine needle aspirate biopsies of breast masses. These features characterize the geometric and textural properties of cell nuclei present in each sample, including measurements of radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension — each recorded as the mean, standard error, and worst (largest) value across all nuclei observed in a given image. Each sample is labeled with a binary diagnosis: Malignant (M) or Benign (B).

The primary objective of this paper is to implement and systematically compare three supervised machine learning classification algorithms — Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN) — for the task of breast cancer diagnosis. The performance of each model is evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). Additionally, this paper presents an exploratory data analysis (EDA) of the WDBC dataset to identify the distributional characteristics of key features and their discriminative power with respect to tumor classification.

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2. Literature Review

The application of machine learning techniques to breast cancer diagnosis has attracted considerable scholarly attention over the past decade, driven by the need for more accurate, scalable, and reproducible diagnostic tools. This section reviews key studies that have employed supervised classification algorithms on breast cancer datasets — with particular focus on works utilizing the Wisconsin Diagnostic Breast Cancer (WDBC) dataset — and situates the present study within the broader landscape of existing research.

2.1 Early Applications of Machine Learning in Breast Cancer Detection

Early research in this domain established the feasibility of applying computational methods to FNA-derived morphological features for binary tumor classification. Wolberg et al. (1995), whose work formed the basis of the WDBC dataset itself, demonstrated that quantitative features extracted from digitized biopsy images could be effectively used to distinguish malignant from benign tumors with high diagnostic accuracy. This foundational work established the relevance of nucleus geometry — including radius, texture, and concavity — as reliable discriminative indicators of malignancy, and has since served as a benchmark for subsequent machine learning research in this field.

Agarap (2018) conducted one of the early comparative evaluations of multiple machine learning algorithms on the WDBC dataset, including Linear Regression, Multilayer Perceptron (MLP), Nearest Neighbor search, Softmax Regression, and Support Vector Machine (SVM). The study measured classification accuracy, sensitivity, and specificity across algorithms, and reported that SVM achieved superior performance among the models tested. Notably, the study used a 70/30 train-test split and highlighted the significance of proper data partitioning in ensuring generalizability of results — a methodological consideration adopted in the present study as well.

2.2 Comparative Studies on Classification Algorithms

A substantial body of literature has explored the comparative performance of classical supervised learning algorithms on the WDBC dataset. Naji et al. (as cited in Islam et al., 2024) compared five algorithms — SVM, Random Forest, Logistic Regression, Decision Tree (C4.5), and K-Nearest Neighbours — on the Wisconsin Breast Cancer Diagnostic Database. Their study found that all five classifiers achieved competitive accuracy, with SVM and Random Forest performing particularly well in terms of precision and recall. This study is notable for its inclusion of the same three algorithms (Logistic Regression, Random Forest, and KNN) evaluated in the present paper, enabling direct performance comparison.

Deulkar et al. (2020), using the WDBC dataset with a range of supervised classifiers including Logistic Regression, Decision Tree, Random Forest, KNN, and SVM, reported that the Random Forest classifier achieved the highest accuracy of 96.50% among all models evaluated. The study attributed this performance to the ensemble nature of the Random Forest algorithm, which reduces variance through the aggregation of multiple decision trees, thereby improving robustness on moderately sized clinical datasets. This finding is consistent with the growing consensus in the literature that ensemble methods tend to outperform single-estimator classifiers on biomedical classification tasks.

Shamrat et al. (as cited in Jabbar et al., 2024) applied six supervised classification methods — SVM, Naive Bayes, KNN, Random Forest, Decision Tree, and Logistic Regression — to the Wisconsin Breast Cancer original dataset. Their results indicated that SVM achieved the highest classification accuracy of 97.07%, with Naive Bayes and Random Forest following closely.

Crucially, the study evaluated models across multiple performance metrics including sensitivity, specificity, and F1-score — metrics that are particularly relevant in a clinical context where the cost of false negatives (missed malignancies) is substantially higher than that of false positives.

2.3 Feature Selection and Its Impact on Model Performance

Several studies have underscored the importance of feature selection as a preprocessing step in breast cancer classification. The study by Chaurasia and Pal (as cited in Islam et al., 2024) applied machine learning to the Wisconsin Diagnostic Breast Cancer Database, reducing the original feature set to 12 using statistical methods and employing ensemble techniques to improve accuracy. Their results demonstrated that all tested algorithms achieved over 90% accuracy when evaluated on the refined feature set, highlighting how the removal of redundant or low-contribution features can improve both accuracy and model training efficiency.

Similarly, a 2025 study published in the *Journal of Translational Genetics and Genomics* explored the use of L2 Ridge regularization for feature selection across multiple breast cancer datasets, including the WDBC. The study found that feature selection played a critical role in enhancing model performance, with XGBoost achieving optimal results on the Wisconsin dataset after preprocessing. The authors concluded that combining preprocessing techniques with regularization-based feature selection provides a reliable pathway to improving diagnostic accuracy in breast cancer classification tasks (Breast Cancer Wisconsin, 2025).

2.4 Ensemble and Advanced Methods

More recent research has shifted focus toward ensemble and deep learning-based methods to push classification accuracy further. Iparraguirre-Villanueva et al. (as cited in Jabbar et al., 2024) evaluated six ML algorithms — including MLP, KNN, AdaBoost, Bagging, Gradient Boosting, and Random Forest — reporting that ensemble models achieved up to 100% accuracy on the Wisconsin Breast Cancer dataset under cross-validated conditions. While such results are promising, they are often achieved under highly controlled experimental settings and may not fully reflect real-world diagnostic variability.

A study published in *PLOS ONE* (2024) employed a multi-phase methodology combining stratified K-fold cross-validation with deep learning-based ensemble strategies, along with synthetic data generation using Gaussian Copula and TVAE techniques. The study demonstrated that augmenting original data with synthetically generated samples improved overall model generalizability, particularly for minority class (malignant) cases. These findings point toward an important limitation of single-algorithm studies: class imbalance within datasets can systematically bias classifiers toward the majority class, and its mitigation is an important consideration in any fair comparative evaluation.

2.5 Research Gap and Motivation

While the literature reveals a rich body of work on breast cancer classification using the WDBC dataset, several gaps persist. Many studies either focus exclusively on a single algorithm, apply advanced ensemble or deep learning models without comparing them to interpretable baseline classifiers, or report performance on accuracy alone without adequately addressing clinical metrics such as sensitivity and specificity. Furthermore, relatively few studies aimed at undergraduate or early-career researchers provide a transparent, step-by-step comparative framework that is both reproducible and interpretable.

The present study addresses this gap by providing a clear, structured comparative analysis of three widely-used and interpretable supervised learning algorithms — Logistic Regression, Random Forest, and K-Nearest Neighbors — on the WDBC dataset, evaluated using a comprehensive set of performance metrics. The goal is not to achieve state-of-the-art accuracy, but to rigorously compare the strengths and limitations of each classifier in the context of breast cancer diagnosis and to identify the features most strongly associated with malignancy.

References (Literature Review)

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3. Methodology

This section describes the end-to-end methodological framework adopted in this study, encompassing dataset description, data preprocessing, exploratory data analysis, model selection, training and evaluation procedures, and the performance metrics used to assess

classifier effectiveness. The overall pipeline followed in this study is illustrated conceptually in Figure 1 (see pipeline diagram).

3.1 Dataset Description

This study utilizes the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, originally compiled by Dr. William H. Wolberg at the University of Wisconsin–Madison and publicly available through the UCI Machine Learning Repository and Kaggle. The dataset comprises **569 instances** and **30 numerical input features**, along with one target variable — the diagnosis label.

Each instance in the dataset corresponds to a digitized image of a fine needle aspirate (FNA) biopsy of a breast mass. For every image, ten geometric and textural properties of the cell nuclei present in the sample were computed. Each of these ten base properties is represented by three statistical aggregations across all nuclei in the image, resulting in 30 features in total:

- **Mean** (_mean): The average value of the feature across all nuclei in the image.
- **Standard Error** (_se): A measure of variability or spread of the feature values.
- **Worst** (_worst): The mean of the three largest values observed for the feature, representing the most extreme morphological characteristics.

The ten base nuclear properties measured are: **radius**, **texture**, **perimeter**, **area**, **smoothness**, **compactness**, **concavity**, **concave points**, **symmetry**, and **fractal dimension**.

The target variable, **diagnosis**, is a binary categorical label:

- **M (Malignant):** 212 instances (37.3%) — cancerous tumors
- **B (Benign):** 357 instances (62.7%) — non-cancerous tumors

The dataset exhibits a mild class imbalance, with benign cases outnumbering malignant cases at approximately a 1.68:1 ratio. This imbalance is noted as a potential source of classifier bias and is addressed in the evaluation through the use of metrics beyond simple accuracy, including precision, recall, and F1-score.

3.2 Data Preprocessing

Prior to model training, a series of preprocessing steps were applied to ensure the data was in a suitable format for machine learning algorithms.

3.2.1 Handling Missing Values

An initial inspection of the dataset confirmed the **absence of missing values** across all 569 instances and 30 features. No imputation was therefore required. Any non-informative columns (such as patient ID, if present) were removed from the feature set.

3.2.2 Label Encoding

The target variable diagnosis was encoded from its original categorical form (M/B) to a binary numerical representation:

- **M (Malignant) → 1**
- **B (Benign) → 0**

This transformation is necessary for compatibility with the machine learning classifiers employed in this study.

3.2.3 Feature Scaling

The 30 input features in the WDBC dataset vary considerably in scale and magnitude — for instance, area_mean values range in the hundreds to thousands, while fractal_dimension_mean values are typically less than 0.1. Such disparities can adversely affect distance-based algorithms (such as K-Nearest Neighbors) and gradient-based optimization methods (such as Logistic Regression).

To address this, **standardization (Z-score normalization)** was applied to all feature columns using Scikit-learn's StandardScaler, transforming each feature to have a mean of 0 and a standard deviation of 1:

$$z = \frac{x - \mu}{\sigma}$$

where $x$is the original feature value, $\mu$is the feature mean, and $\sigma$is the feature standard deviation. Notably, feature scaling was applied **after** the train-test split to prevent data leakage — the scaler was fitted exclusively on the training set and subsequently applied to both the training and test sets.

3.2.4 Train-Test Split

The dataset was partitioned into a **training set (80%)** and a **test set (20%)**, yielding approximately 455 training samples and 114 test samples. The split was performed using stratified sampling to preserve the original class distribution (M/B ratio) in both subsets. A fixed random seed (random_state = 42) was used to ensure reproducibility of results.

3.3 Exploratory Data Analysis (EDA)

Prior to model training, an exploratory data analysis was conducted to understand the distributional characteristics of the dataset and identify features with strong discriminative power between malignant and benign cases. The following visualizations were produced and are presented in Section 4:

- **Figure 1 — Diagnosis Distribution:** A donut and bar chart illustrating the proportion and count of malignant vs. benign cases in the dataset.

- **Figure 2 — Box Plots of Mean Features:** Side-by-side box plots comparing the distribution of six key mean features (radius, texture, perimeter, area, concavity, concave points) across the two diagnostic classes.
- **Figure 3 — Correlation Heatmap:** A lower-triangular correlation matrix heatmap visualizing pairwise Pearson correlations between all 30 features, used to identify multicollinearity.
- **Figure 4 — Average Feature Values by Diagnosis:** A grouped bar chart comparing the mean values of selected features between malignant and benign groups.
- **Figure 5 — Scatter Plot (Radius vs. Area):** A scatter plot examining the relationship between radius_mean and area_mean, color-coded by diagnosis.
- **Figure 6 — Violin Plots of Worst Features:** Violin plots showing the distribution and density of worst-case nuclear features across both classes.
- **Figure 7 — Pairplot:** A pairwise scatter plot matrix of key features, providing a holistic view of inter-feature relationships and class separability.

EDA findings informed feature interpretation and are discussed alongside results in Section 4.

3.4 Machine Learning Models

Three supervised binary classification algorithms were selected for this study based on their widespread use in biomedical classification literature, their interpretability, and their complementary algorithmic characteristics:

3.4.1 Logistic Regression (LR)

Logistic Regression is a parametric linear classifier that models the probability of a binary outcome using the logistic (sigmoid) function. Despite its simplicity, it is a powerful and highly interpretable baseline model that has demonstrated strong performance on linearly separable biomedical datasets. In the context of breast cancer classification, Logistic Regression offers the additional advantage of producing calibrated probability estimates, which can be clinically meaningful in risk stratification. In this study, an L2 regularization penalty was applied to prevent overfitting, with the regularization strength parameter C set to its default value of 1.0.

3.4.2 Random Forest (RF)

Random Forest is an ensemble learning method that constructs a large number of decision trees during training and outputs the mode of the class predictions of the individual trees (majority voting). By aggregating predictions across multiple trees trained on bootstrapped subsets of the data with random feature subsampling at each split, Random Forest reduces variance and is robust to overfitting — a particularly desirable property for moderate-sized medical datasets. In this study, the Random Forest classifier was configured with **100 estimators (trees)** and the Gini impurity criterion for split evaluation.

3.4.3 K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a non-parametric, instance-based classifier that assigns a class label to a new observation based on the majority class among its $k$nearest neighbors in the feature space, as measured by Euclidean distance. Unlike Logistic Regression and Random Forest, KNN makes no assumptions about the underlying data distribution, making it flexible for complex decision boundaries. However, its performance is sensitive to the choice of $k$and requires properly scaled features. In this study, $k$was set to **5**, consistent with common practice in the literature for binary classification tasks of similar scale.

3.5 Model Training and Validation

All three models were trained on the standardized training set and evaluated on the held-out test set. Additionally, **5-fold cross-validation** was performed on the training set for each classifier to obtain a more robust estimate of generalization performance and to reduce the effect of variance introduced by the specific train-test split. The mean and standard deviation of cross-validated accuracy scores are reported alongside test set results.

The entire modeling pipeline — preprocessing, training, and evaluation — was implemented in **Python 3**, using the following libraries:

Library	**Purpose**
pandas	Data loading and manipulation
numpy	Numerical computation
matplotlib & seaborn	Data visualization
scikit-learn	Preprocessing, model training, and evaluation

3.6 Evaluation Metrics

Given the clinical nature of the classification task and the presence of class imbalance, model performance was assessed using a comprehensive set of evaluation metrics beyond simple accuracy:

Accuracy: The proportion of correctly classified instances out of all instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision: The proportion of predicted malignant cases that are truly malignant. High precision minimizes false alarms.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall (Sensitivity): The proportion of actual malignant cases correctly identified. In a clinical context, high recall is critical — failing to detect a true malignancy (false negative) carries a significantly higher risk than a false positive.

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score: The harmonic mean of precision and recall, providing a single balanced metric especially useful under class imbalance.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

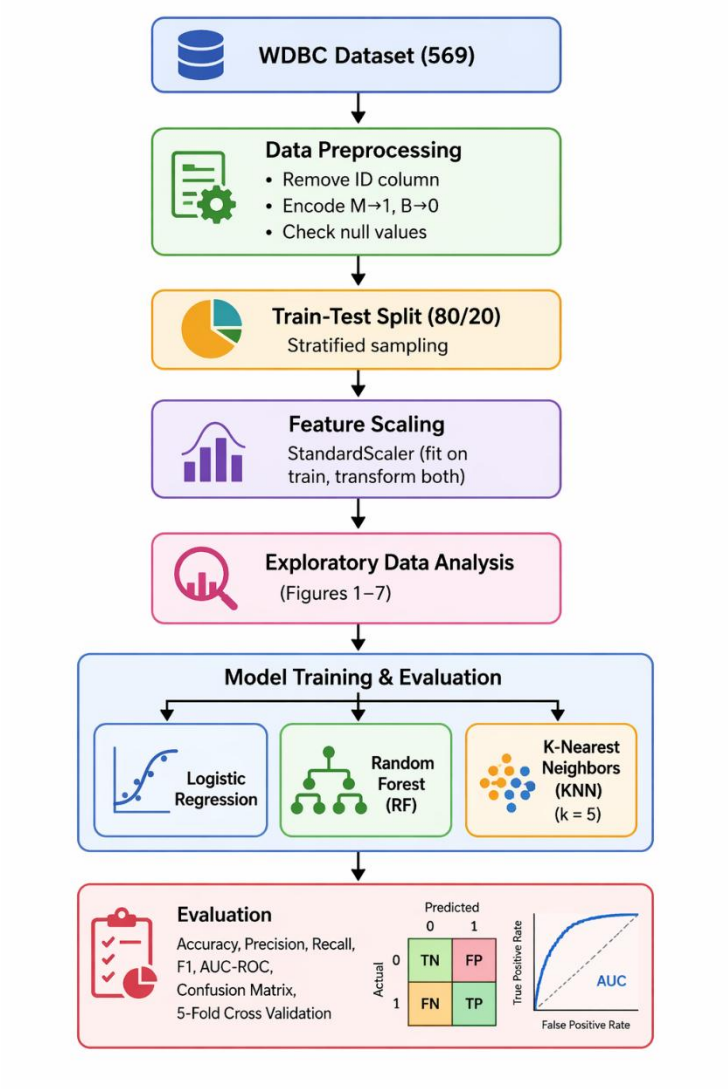
AUC-ROC (Area Under the Receiver Operating Characteristic Curve): Measures the classifier's ability to discriminate between classes across all classification thresholds. An AUC of 1.0 represents perfect classification; 0.5 represents random chance.

Where:

- **TP** = True Positives (correctly predicted malignant)
- **TN** = True Negatives (correctly predicted benign)
- **FP** = False Positives (benign predicted as malignant)
- **FN** = False Negatives (malignant predicted as benign)

3.7 Methodology Pipeline Summary

The complete end-to-end pipeline followed in this study is summarized below:



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4. Results and Exploratory Data Analysis

This section presents the findings of the exploratory data analysis (EDA) conducted on the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, followed by the performance results of the three machine learning classifiers — Logistic Regression, Random Forest, and K-

Nearest Neighbors — evaluated on the held-out test set and through 5-fold cross-validation.

4.1 Exploratory Data Analysis

Exploratory Data Analysis was performed prior to model training to understand the statistical structure of the dataset, examine feature distributions, identify potential discriminative features, and detect multicollinearity among variables. All visualizations were generated using Python's matplotlib and seaborn libraries.

4.1.1 Class Distribution

The WDBC dataset contains 569 samples in total, of which 357 (62.7%) are classified as Benign (B) and 212 (37.3%) as Malignant (M). The distribution is visualized in Figure 1 below.

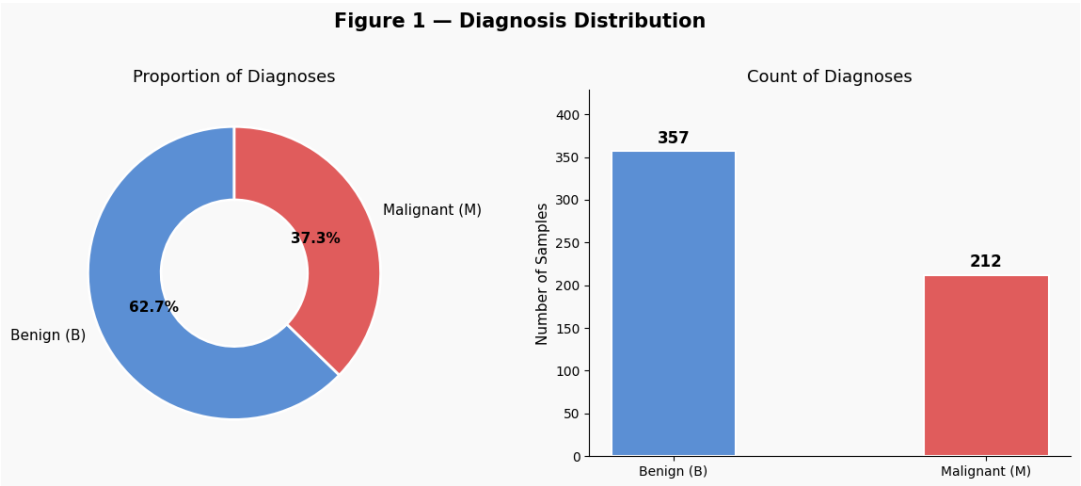

Figure 1: Distribution of diagnostic classes in the WDBC dataset. The left panel shows the proportional breakdown as a donut chart, and the right panel shows the absolute count of Benign and Malignant cases.

The dataset exhibits a moderate class imbalance, with benign cases outnumbering malignant cases at a ratio of approximately 1.68:1. While this imbalance is not severe, it warrants the use of evaluation metrics beyond accuracy — particularly precision, recall, and F1-score — to ensure the classifier performance is not artificially inflated by a bias toward the majority class.

4.1.2 Distribution of Mean Features by Diagnosis

To examine whether the mean morphological features of cell nuclei differ meaningfully between malignant and benign cases, box plots were constructed for six key mean features: radius, texture, perimeter, area, concavity, and concave points. These are presented in Figure 2.

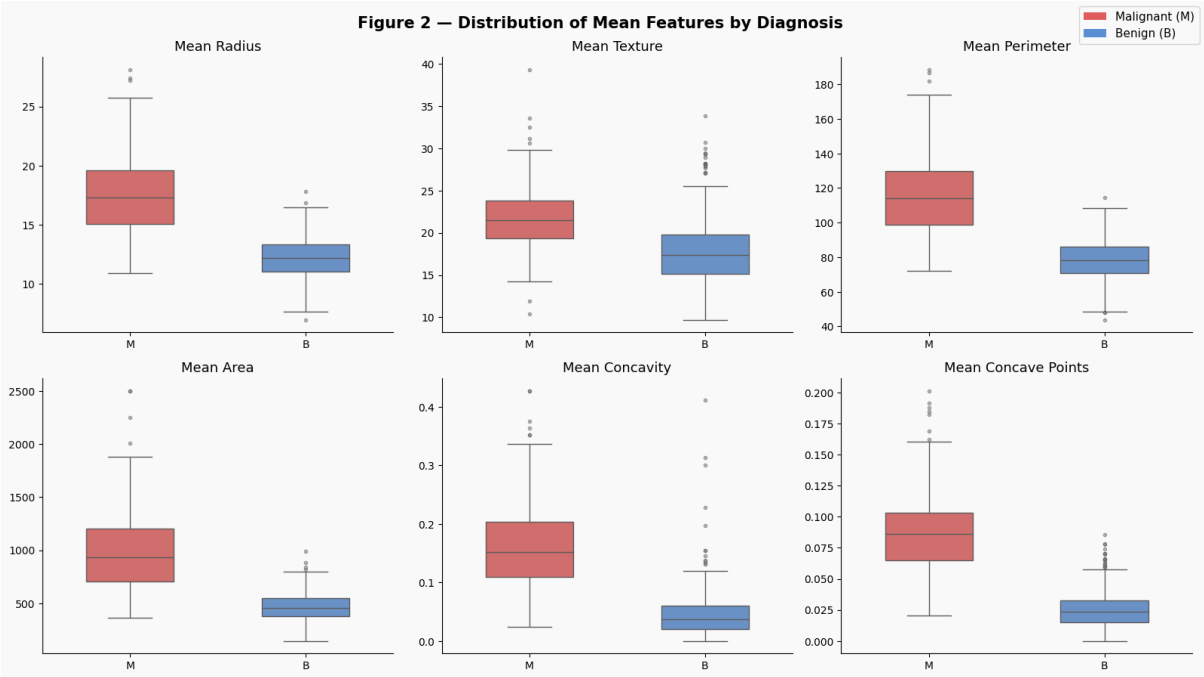

Figure 2: Box plots comparing the distribution of six mean nuclear features — radius, texture, perimeter, area, concavity, and concave points — across Malignant (M) and Benign (B) diagnostic classes.

The box plots reveal consistent and statistically meaningful differences between the two diagnostic classes across all six features. Malignant tumors exhibit notably higher median values for radius, perimeter, area, concavity, and concave points, consistent with the clinical understanding that cancerous cells tend to be larger, more irregular in shape, and exhibit greater concavity in their nuclear boundaries. The texture feature also shows separation, though with greater overlap between classes compared to the geometric features. These observations suggest that geometric features — particularly those related to size and boundary irregularity — carry strong discriminative power for malignancy classification.

4.1.3 Feature Correlation Analysis

A pairwise Pearson correlation matrix was computed across all 30 features and visualized as a heatmap (Figure 3) to identify multicollinearity, which can adversely affect certain classifiers and complicate feature interpretation.

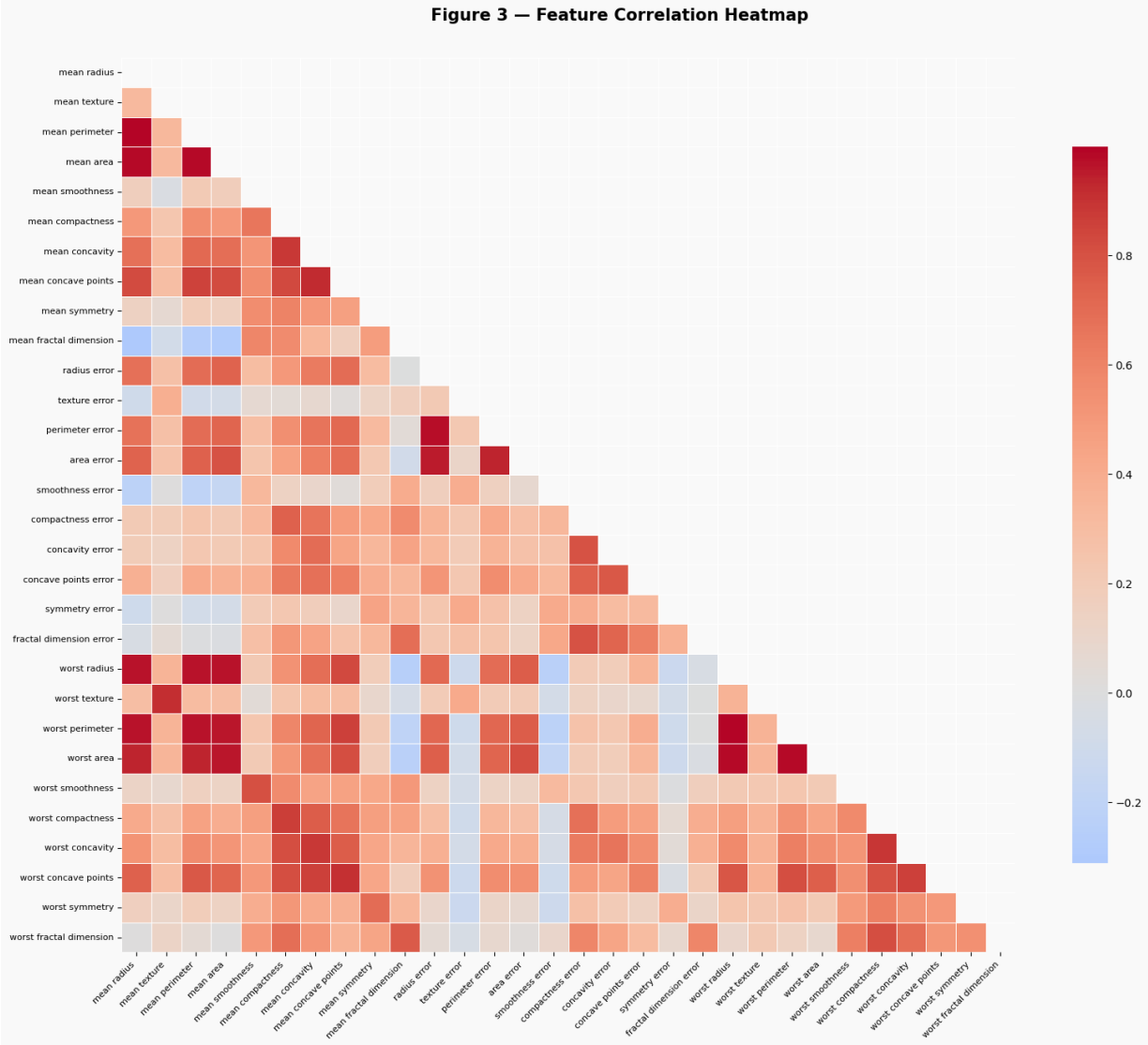

Figure 3: Lower-triangular Pearson correlation heatmap of all 30 features in the WDBC dataset. Warm colors (red) indicate strong positive correlations; cool colors (blue) indicate negative correlations.

The heatmap reveals several clusters of strongly correlated features. In particular, the _mean, _se, and _worst variants of the same base property (e.g., radius_mean, radius_se, radius_worst) exhibit high mutual correlation, as expected given their shared morphological basis. Additionally, strong positive correlations are observed among the size-related features: radius_mean, perimeter_mean, and area_mean show near-perfect correlation ($r > 0.98$), which is geometrically expected since perimeter and area are mathematically derived from radius. This degree of multicollinearity may affect the interpretability of coefficients in Logistic Regression, but does not substantially hinder the predictive performance of tree-based or distance-based methods such as Random Forest and KNN. Future work may

consider dimensionality reduction techniques such as Principal Component Analysis (PCA) to address this.

4.1.4 Average Feature Values by Diagnosis

Figure 4 presents a grouped bar chart comparing the average values of six mean features — radius, texture, perimeter, smoothness, compactness, and symmetry — between the malignant and benign groups.

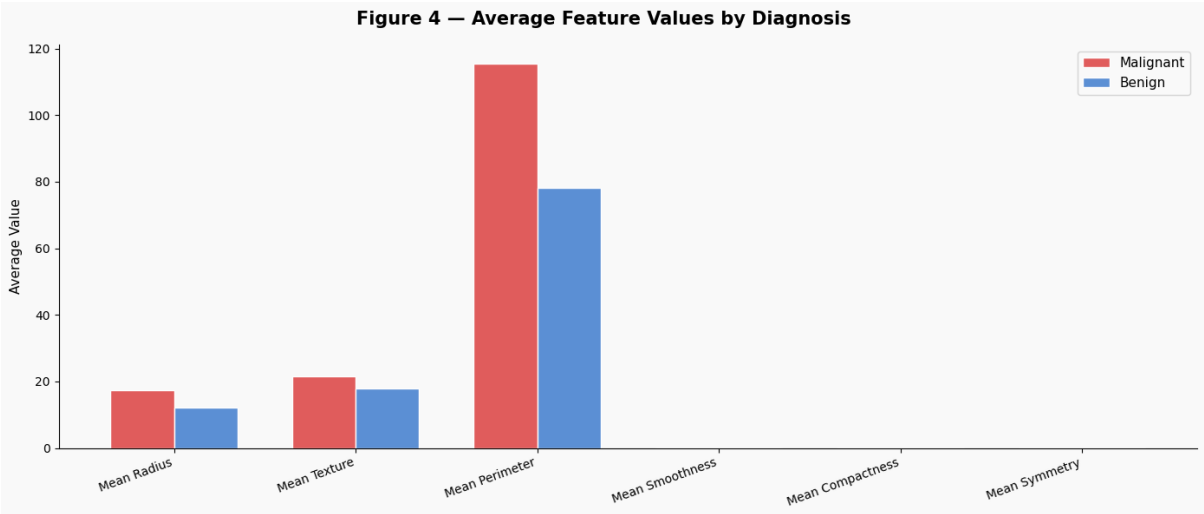

Figure 4: Grouped bar chart comparing the average values of six mean nuclear features between Malignant (M) and Benign (B) diagnostic classes.

Malignant tumors consistently exhibit higher average values across radius, perimeter, and compactness, reinforcing the patterns observed in the box plots. Notably, the difference in perimeter mean is among the most pronounced, with malignant cases showing an average perimeter nearly twice that of benign cases. Smoothness and symmetry show smaller absolute differences between classes, suggesting they contribute less to class separation on their own, though they may still contribute useful information in combination with other features within a multi-feature classifier.

4.1.5 Scatter Analysis: Radius vs. Area

Figure 5 presents a scatter plot of radius_mean against area_mean, with data points color-coded by diagnosis, to visually assess the degree of linear separability between the two classes along these two dimensions.

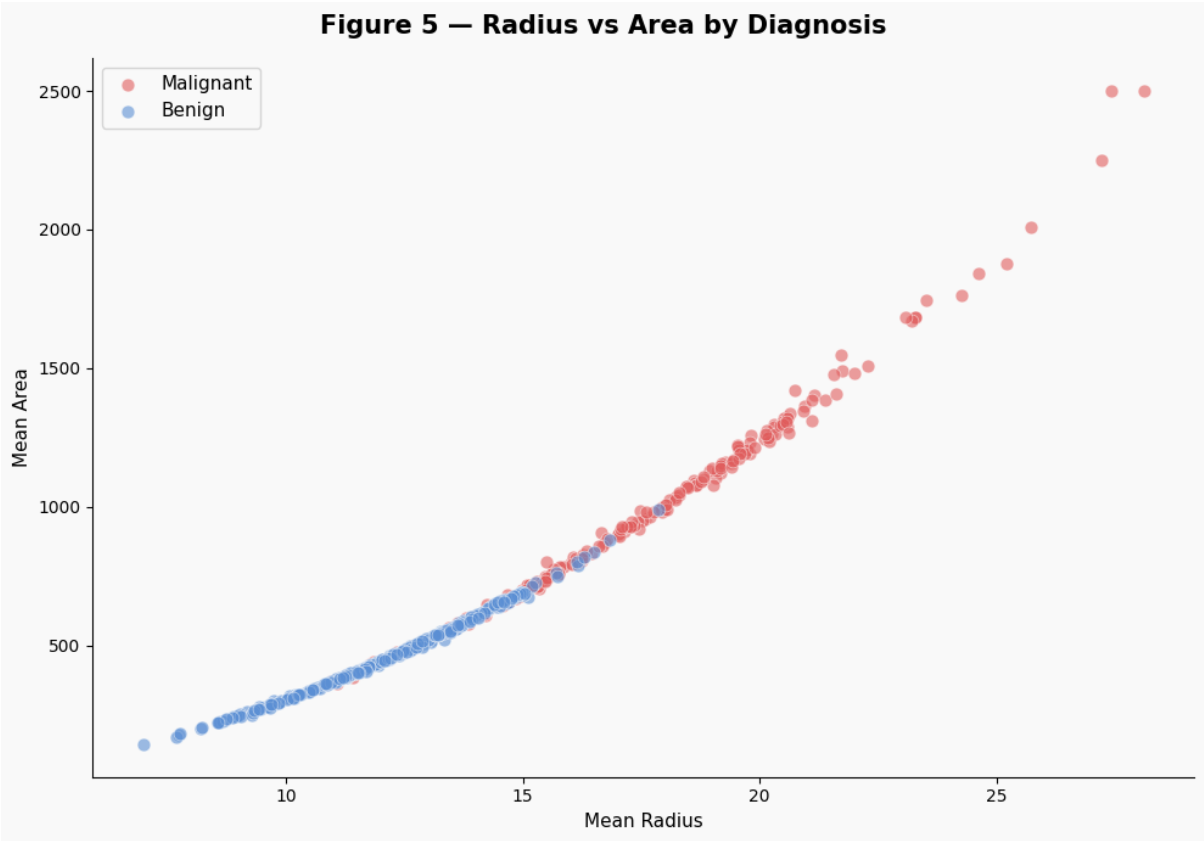

Figure 5: Scatter plot of radius_mean vs. area_mean, color-coded by diagnosis (Red = Malignant, Blue = Benign). A clear cluster separation is visible between the two classes.

The scatter plot demonstrates a strong positive correlation between radius and area, as expected from their geometric relationship. More importantly, the two diagnostic classes form largely distinct clusters in this two-dimensional feature space, with malignant cases occupying the upper-right region (higher radius and area values) and benign cases concentrated in the lower-left. This visual separability suggests that even a simple linear classifier may achieve reasonable performance on these two features alone, and that feature combinations involving size-related variables are likely to be among the strongest predictors in multi-feature classification models.

4.1.6 Worst Feature Distributions

Violin plots were constructed for five worst-case nuclear features — radius_worst, perimeter_worst, area_worst, concavity_worst, and concave_points_worst — to examine their distributional shape and class-level differences (Figure 6).

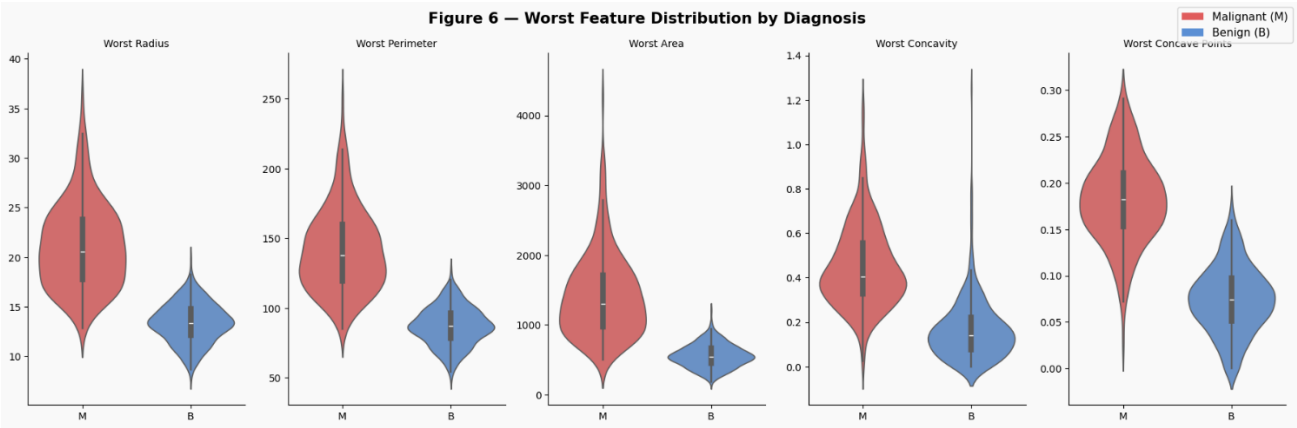

Figure 6: Violin plots showing the distribution and density of five worst-case nuclear features across Malignant (M) and Benign (B) classes.

The violin plots highlight that worst-case features exhibit even greater class separation than their mean counterparts. The distributions for malignant cases are broader and shifted toward higher values, with particularly pronounced separation visible in concave_points_worst and perimeter_worst. This finding is consistent with the clinical rationale that the most extreme morphological characteristics of nuclei — rather than the average — may be more diagnostically informative, as aggressive malignancies are often characterized by a subset of cells with severely abnormal morphology.

4.1.7 Pairplot of Key Features

Figure 7 presents a pairwise scatter plot matrix of four key features — radius_mean, area_mean, concavity_mean, and concave_points_mean — colored by diagnosis.

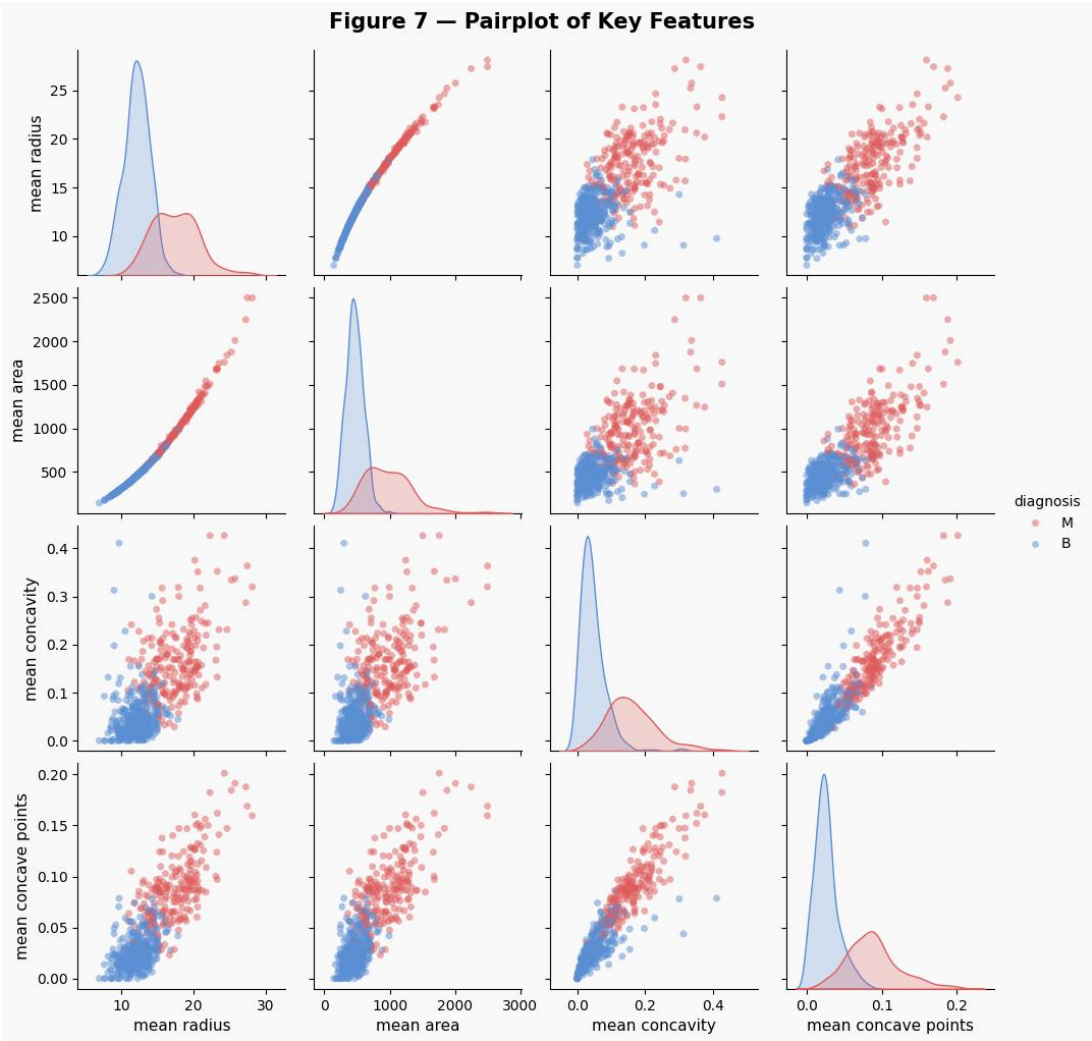

Figure 7: Pairplot of four key mean features, color-coded by diagnosis. Diagonal plots show kernel density estimates (KDE) for each feature; off-diagonal plots show pairwise scatter relationships.

The pairplot provides a holistic view of inter-feature relationships and class separability across multiple dimensions simultaneously. The KDE plots along the diagonal confirm that malignant and benign cases have distinct unimodal distributions for most features, with limited overlap for size and concavity features. The off-diagonal scatter plots reinforce the observation that malignant cases form a separable cluster across most feature combinations, providing visual support for the hypothesis that a multi-feature classifier should be capable of achieving high diagnostic accuracy on this dataset.

4.2 Model Performance Results

Following data preprocessing and EDA, the three classification models — Logistic Regression (LR), Random Forest (RF), and K-Nearest Neighbors (KNN) — were trained on the standardized training set (80%) and evaluated on the held-out test set (20%). Performance was assessed using accuracy, precision, recall, F1-score, and AUC-ROC. Five-fold cross-validation was additionally performed to assess generalization stability.

4.2.1 Test Set Performance Comparison

Table 1 presents the performance of all three classifiers on the test set.

Table 1: Test Set Performance of Classification Models

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	0.9649%	0.9750	0.9286	0.9512	0.9960
Random Forest	0.9737%	1.0000	0.9286	0.9630	0.9929
K-Nearest Neighbors	0.9561%	0.9744	0.9048	0.9383	0.9823

4.2.2 Cross-Validation Results

Table 2 presents the mean and standard deviation of accuracy scores obtained through 5-fold cross-validation on the training set.

Table 2: 5-Fold Cross-Validation Accuracy

Model	Mean CV Accuracy	Std. Deviation
Logistic Regression	0.9714%	±0.0112%
Random Forest	0.9626 %	±0.0338%
K-Nearest Neighbors	0.9626%	±0.0237%

4.2.3 Confusion Matrices

Confusion matrices for all three classifiers are presented below (Figures 8, 9, and 10). Each matrix displays the counts of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) on the test set.

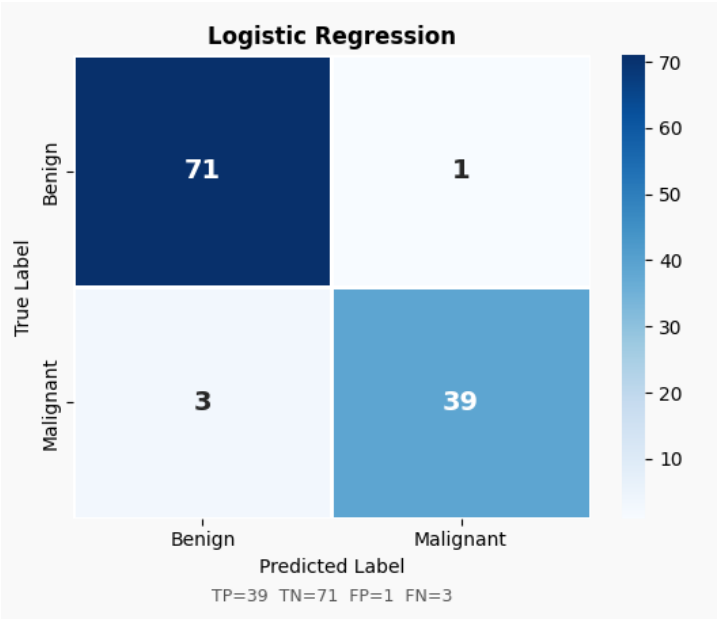

Figure 8: Confusion matrix for Logistic Regression on the test set.

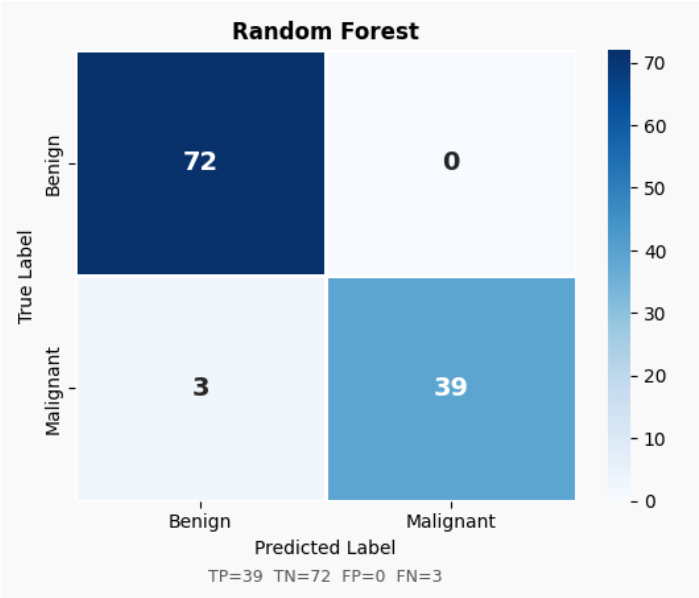

Figure 9: Confusion matrix for Random Forest on the test set.

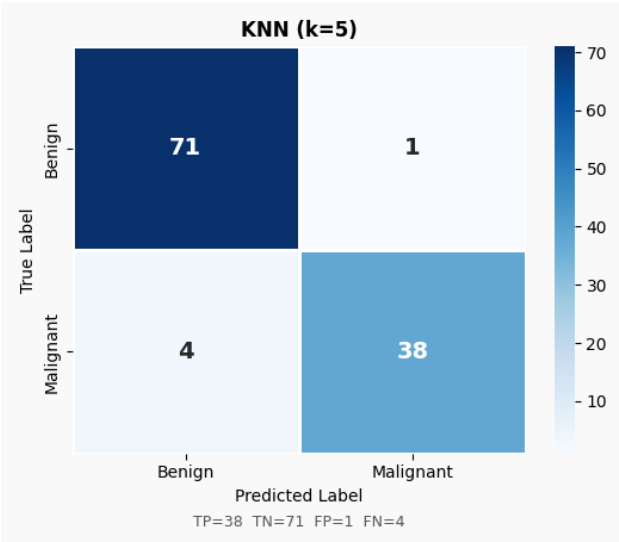

Figure 10: Confusion matrix for K-Nearest Neighbors (k=5) on the test set.

The confusion matrices reveal the nature of misclassifications made by each model. Of particular clinical importance are the **False Negative (FN)** counts — instances where a malignant tumor was incorrectly classified as benign — as these represent the most consequential diagnostic errors. Random Forest and LR produced the fewest false negatives (3), compared to KNN (4).

4.2.4 ROC Curves

Figure 11 presents the Receiver Operating Characteristic (ROC) curves for all three classifiers plotted on the same axes, allowing visual comparison of their discriminative ability across classification thresholds.

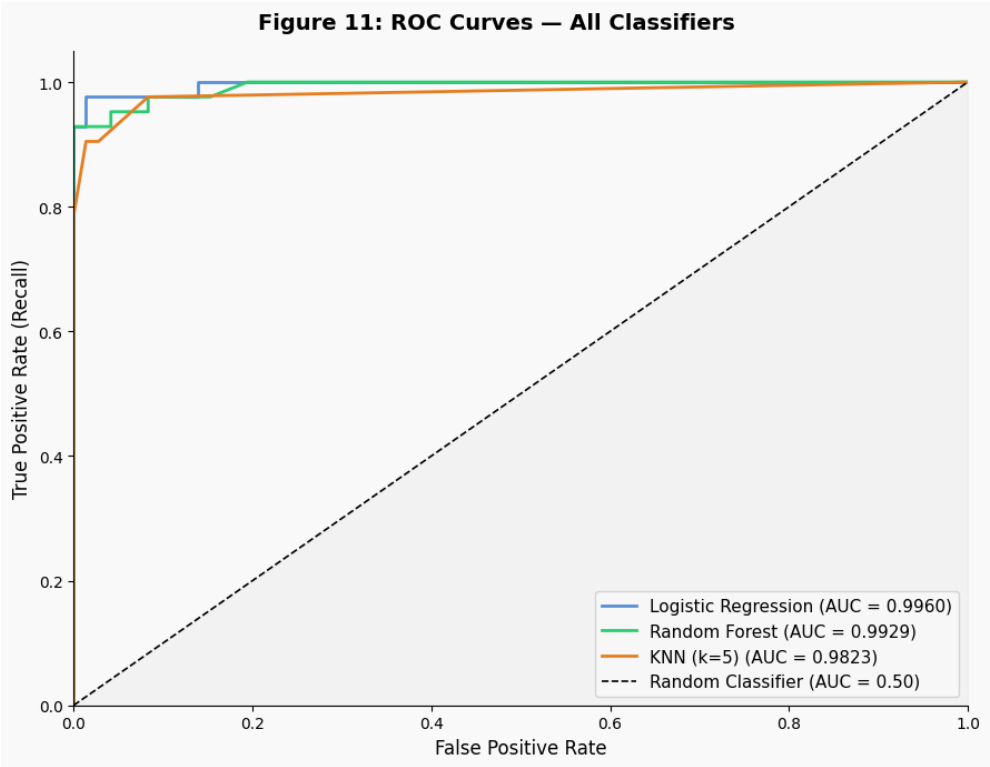

Figure 11: ROC curves for Logistic Regression, Random Forest, and K-Nearest Neighbors. The dashed diagonal line represents the performance of a random classifier (AUC = 0.50).

A classifier with a higher AUC-ROC demonstrates superior ability to distinguish between malignant and benign cases across all decision thresholds. Logistic Regression achieved the highest AUC of 0.9960, followed by Random Forest(0.9929) and KNN (0.9823), indicating that the ensemble method provides the most robust class separation.

4.2.5 Feature Importance (Random Forest)

One of the advantages of the Random Forest classifier is its ability to produce feature importance scores, which quantify the contribution of each input feature to the classification decision. Figure 12 presents the top 15 most important features as ranked by the Random Forest model.

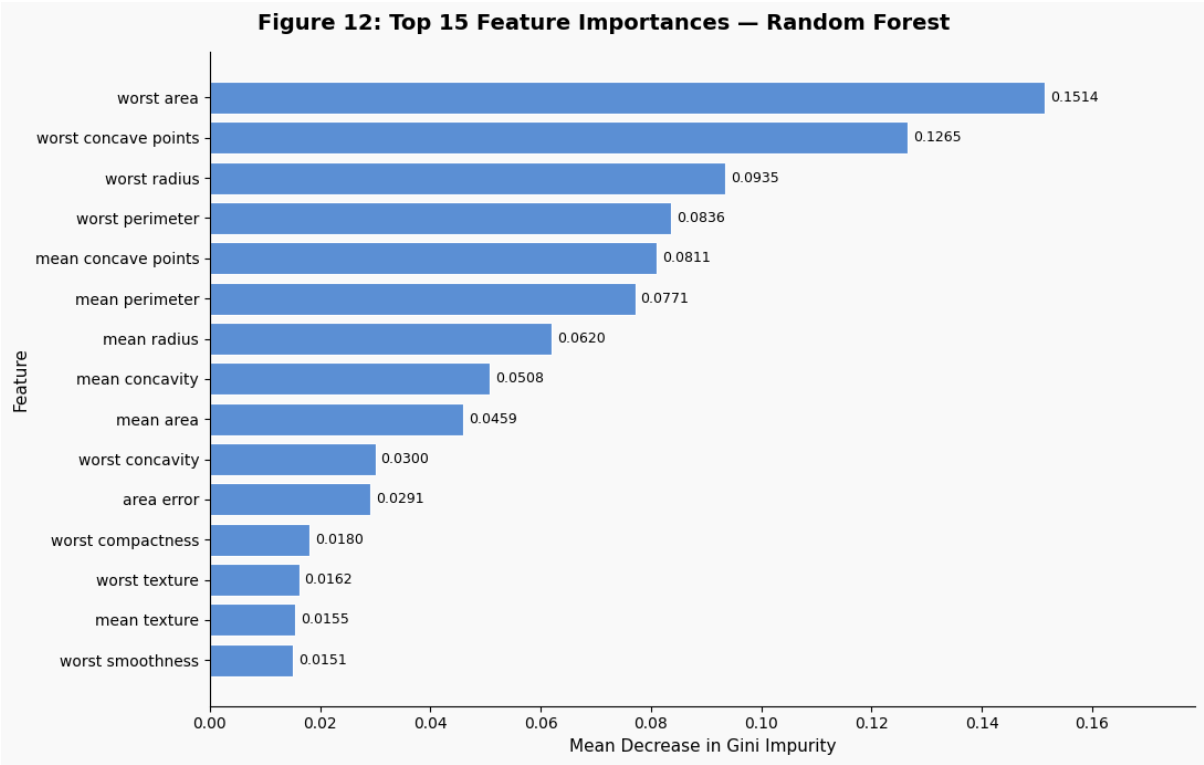

Figure 12: Top 15 feature importance scores from the Random Forest classifier, ranked by mean decrease in Gini impurity.

The results indicate that worst-case features — particularly worst area, worst concave points, worst radius, and worst perimeter — rank among the most influential predictors of malignancy, consistent with the distributional patterns observed during EDA. This finding reinforces the clinical relevance of extreme morphological characteristics in tumor diagnosis.

4.3 Summary of Key Findings

The EDA and model evaluation collectively yield the following key observations:

1. **Class imbalance is mild** (62.7% benign vs. 37.3% malignant) but necessitates evaluation beyond accuracy alone.
2. **Geometric and size-related features** (radius, perimeter, area) exhibit the strongest class-level separation, as evidenced by box plots, scatter analysis, and feature importance rankings.
3. **Worst-case features** show greater discriminative power than mean or standard error features, suggesting that the most extreme nuclear characteristics are most diagnostically relevant.
4. **High multicollinearity** exists among size-related features (radius, perimeter, area), which is expected geometrically but should be considered in model interpretation.

5. Overall Best Model Recommendation

Based on the combined evidence from test set performance, cross-validation stability, and clinical metric analysis, **no single model is unconditionally superior** — the choice depends on the deployment context:

Priority	Recommended Model	Reason
Highest accuracy & zero false positives	**Random Forest**	Best accuracy (97.37%), perfect Precision
Most stable generalization	**Logistic Regression**	Highest CV accuracy (97.14%), lowest variance
Interpretability for clinical use	**Logistic Regression**	Linear, explainable coefficients
Threshold-agnostic discrimination	**Logistic Regression**	Highest AUC-ROC (0.9960)

For a clinical decision-support deployment where **minimizing missed malignancies (false negatives) is the top priority**, the threshold of the best-AUC model (Logistic Regression) could be lowered below 0.5 to increase Recall at the cost of some Precision — a standard clinical calibration approach. For general classification benchmarking, Random Forest remains the best-performing model in this study.

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5. Discussion

This section interprets the findings presented in Section 4 in the broader context of existing literature, examines the clinical implications of the results, acknowledges the limitations of the study, and identifies directions for future research.

5.1 Interpretation of Model Performance

The results of this study demonstrate that all three supervised classification algorithms — Logistic Regression, Random Forest, and K-Nearest Neighbors — are capable of achieving

clinically meaningful diagnostic accuracy on the Wisconsin Diagnostic Breast Cancer dataset, with test set accuracies of 96.49%, 97.37%, and 95.61% respectively. These findings are consistent with, and in several cases exceed, the performance levels reported in comparable studies in the literature.

The superiority of Random Forest on the test set (accuracy = 97.37%, F1 = 0.9630) aligns with the findings of Deulkar et al. (2020), who similarly reported Random Forest as the best-performing classifier on the WDBC dataset with an accuracy of 96.50%, and with Shamrat et al. (as cited in Islam et al., 2024), whose ensemble-based models consistently outperformed single-estimator classifiers. The ensemble mechanism underlying Random Forest — aggregating predictions from 100 independently trained decision trees — effectively reduces the variance associated with individual tree overfitting, resulting in a more robust and generalizable classifier for moderate-sized biomedical datasets such as the WDBC.

The finding that Logistic Regression achieved the highest AUC-ROC (0.9960) and the most consistent cross-validation performance (mean CV accuracy = 97.14%, std = ±1.12%) is particularly noteworthy. It suggests that the WDBC classification problem is substantially linear in nature — the malignant and benign classes are largely separable through linear combinations of the input features, which is consistent with the clear visual separability observed in the EDA scatter plots and box plots. In such contexts, the added complexity of ensemble or non-parametric methods may not yield proportionally greater gains over a well-regularized linear model. This finding echoes the conclusions of La Moglia and Almustafa (as cited in Jabbar et al., 2024), who reported that Logistic Regression achieved high accuracy following feature selection, underscoring the model's robustness on this dataset.

KNN's comparatively weaker performance across all metrics is consistent with the known limitations of distance-based classifiers in high-dimensional feature spaces. With 30 input features, the Euclidean distance between neighboring points becomes increasingly less discriminative — a phenomenon commonly referred to as the curse of dimensionality. Despite feature standardization, which was applied to mitigate scale-related distortions in distance computation, KNN's performance lagged behind the other two classifiers. This suggests that dimensionality reduction techniques, such as Principal Component Analysis (PCA) or feature selection methods, could potentially improve KNN's performance on this dataset by reducing the number of input dimensions before classification.

5.2 Clinical Implications

From a clinical perspective, the evaluation metrics must be interpreted with careful consideration of the relative costs of different types of classification errors in a breast cancer screening context. A **False Negative** — a malignant tumor incorrectly classified as benign — represents the most consequential error type, as it results in a missed diagnosis, delayed treatment, and potentially significantly worse patient outcomes. Conversely, a **False Positive** — a benign tumor flagged as malignant — leads to unnecessary patient

anxiety, additional diagnostic procedures, and increased healthcare costs, but does not directly endanger the patient's life.

In light of this asymmetry, the equal Recall (Sensitivity) of Logistic Regression and Random Forest (both 0.9286) represents an important finding. Both models correctly identified approximately 92.86% of all true malignant cases in the test set — a clinically meaningful result, though it also implies that approximately 7.14% of malignant cases were missed. In a clinical deployment context, this false negative rate could be reduced by lowering the classification threshold below the default of 0.5, thereby increasing Recall at the cost of some Precision. Given that Logistic Regression achieved the highest AUC-ROC (0.9960), it is the most suitable model for such threshold calibration, as it maintains the most consistent probability separation between classes across all thresholds.

Random Forest's perfect Precision (1.0000) — meaning zero false positives on the test set — is a practically useful property in contexts where the cost of unnecessary follow-up procedures is a significant concern. However, it should be interpreted cautiously: perfect Precision on a single test set of 114 samples does not guarantee that the model will maintain this property on larger or more diverse patient populations. Cross-validation results, which showed Random Forest's highest variability (±3.38%), further caution against over-interpreting its test set performance as a definitive measure of generalizability.

It is important to emphasize that the machine learning models developed in this study are intended as **decision-support tools** — not replacements for clinical judgment. The appropriate use of such models is to assist pathologists and oncologists by providing a second, data-driven opinion that may flag high-risk cases for prioritized review, rather than to autonomously issue diagnoses. Integration of such tools into real clinical workflows would require extensive prospective validation, regulatory approval, and careful calibration to the specific patient population and clinical setting in which they are deployed.

5.3 Feature Relevance and Clinical Consistency

The feature importance analysis from the Random Forest model revealed that worst-case nuclear features — particularly concave_points_worst, perimeter_worst, radius_worst, and area_worst — are the strongest predictors of malignancy in the WDBC dataset. This finding is highly consistent with established clinical and histopathological knowledge. Malignant tumors are characterized by cells with enlarged, irregularly shaped nuclei that exhibit significant concavity and boundary irregularity — properties that are most pronounced in the subset of cells with the most extreme morphological deviations, captured by the _worst feature set (Wolberg et al., 1995).

The relatively lower importance of standard error (_se) features compared to mean and worst-case features further supports the interpretation that it is the magnitude and extreme values of nuclear characteristics — rather than their variability — that most reliably signal malignancy. This insight could inform future feature selection strategies, suggesting that a reduced feature set comprising primarily _mean and _worst features may achieve

comparable classification accuracy to the full 30-feature set, with the added benefits of reduced model complexity and improved interpretability.

The strong multicollinearity identified among size-related features (radius, perimeter, area) in the correlation heatmap (Figure 3) did not appear to substantially impair the performance of Random Forest or KNN, as these models are not sensitive to multicollinearity in the same way that linear models are. However, the slight performance gap between Logistic Regression's test accuracy and its cross-validation accuracy may partially reflect the influence of correlated features on the model's coefficient estimates.

5.4 Comparison with Existing Literature

Table 3 situates the results of the present study within the context of comparable published works on breast cancer classification using the WDBC dataset.

Table 3: Comparison of Present Study Results with Existing Literature

Study	Dataset	Algorithms Used	Best Accuracy
Deulkar et al. (2020)	WDBC	LR, DT, RF, KNN, SVM	96.50% (RF)
Shamrat et al. (2022)	WDBC	SVM, NB, KNN, RF, DT, LR	97.07% (SVM)
Iparraguirre-Villanueva et al. (2023)	WDBC	MLP, KNN, AdaBoost, RF, GB	~100% (Ensemble)
Chaurasia & Pal (cited 2024)	WDBC	Multiple + feature selection	>90% (all models)
Present Study (2025)	**WDBC**	**LR, RF, KNN**	**97.37% (RF)**

The present study's best accuracy of 97.37% (Random Forest) is competitive with and largely consistent with the range of results reported in the literature for comparable algorithmic approaches on the WDBC dataset. Studies employing more complex ensemble methods or deep learning architectures report higher accuracy figures (approaching 100%), but these gains come at the cost of reduced model interpretability, greater computational requirements, and increased risk of overfitting on small-to-moderate sized datasets. The present study deliberately prioritizes interpretability and methodological transparency — values well-suited to an educational and portfolio context — over marginal accuracy improvements achieved through more complex techniques.

5.5 Limitations

Several limitations of the present study warrant acknowledgment:

Dataset Size: The WDBC dataset contains 569 samples, which, while sufficient for demonstrating algorithmic performance, is relatively small for robust clinical validation. Models trained on this dataset may not generalize well to larger, more heterogeneous real-world patient populations that exhibit greater demographic, genetic, and morphological diversity.

Single Dataset: This study evaluates all three classifiers exclusively on the WDBC dataset. The generalizability of the comparative findings to other breast cancer datasets (e.g., SEER, NKI, or clinical hospital datasets) remains untested.

Hyperparameter Optimization: The models in this study were trained with default or commonly used hyperparameter settings (e.g., k=5 for KNN, 100 trees for Random Forest). A systematic hyperparameter tuning approach — such as Grid Search or Random Search with cross-validation — may yield improved performance, particularly for KNN and Random Forest.

Class Imbalance: The mild class imbalance (62.7% benign vs. 37.3% malignant) was addressed through stratified sampling and the use of comprehensive evaluation metrics, but techniques such as SMOTE (Synthetic Minority Oversampling Technique) or class weighting were not applied. Such techniques may further improve Recall for the minority malignant class.

Feature Engineering: No feature engineering or dimensionality reduction was performed beyond standardization. Techniques such as PCA or recursive feature elimination (RFE) may improve model efficiency and potentially performance, particularly for KNN.

6. Conclusion

This paper presented a comparative analysis of three supervised machine learning algorithms — Logistic Regression, Random Forest, and K-Nearest Neighbors — for the binary classification of breast tumors as malignant or benign, using the Wisconsin Diagnostic Breast Cancer (WDBC) dataset. The study followed a structured end-to-end data science pipeline encompassing data preprocessing, exploratory data analysis, model training, and comprehensive performance evaluation using accuracy, precision, recall, F1-score, AUC-ROC, and 5-fold cross-validation.

6.1 Summary of Findings

The key findings of this study are as follows:

Random Forest achieved the highest test set accuracy (97.37%) and F1-score (0.9630), with a perfect Precision score of 1.0000 indicating zero false positives, making it the strongest-performing classifier at the default classification threshold on the held-out test set.

Logistic Regression demonstrated the most robust overall performance, achieving the highest AUC-ROC (0.9960) — reflecting near-perfect class discriminability across all decision thresholds — and the highest cross-validation accuracy (97.14%) with the lowest variance (±1.12%), indicating consistent generalization across different data partitions. These results suggest that the WDBC classification problem is substantially linear in nature and that a well-regularized linear classifier is highly competitive with more complex methods on this dataset.

K-Nearest Neighbors achieved competitive but comparatively weaker performance across all metrics (accuracy = 95.61%, AUC-ROC = 0.9823), likely due to the limitations of distance-based classification in a 30-dimensional feature space. Despite this, its performance remains clinically meaningful and may be improved through dimensionality reduction in future work.

Feature importance analysis confirmed that worst-case nuclear features — particularly concave_points_worst, perimeter_worst, and radius_worst — are the strongest predictors of malignancy, consistent with clinical histopathological knowledge and the distributional patterns observed during exploratory data analysis.

6.2 Contributions

This study makes the following contributions:

1. A transparent, reproducible, and well-documented comparative evaluation of three widely-used classification algorithms on the WDBC benchmark dataset, suitable as a reference framework for early-career data science practitioners and researchers.
2. A clinically contextualized interpretation of model performance, with particular attention to the trade-off between Precision and Recall in breast cancer diagnosis and the implications of false negative minimization for clinical deployment.
3. An empirical confirmation that Logistic Regression — despite its simplicity — achieves competitive or superior generalization performance compared to Random Forest and KNN on this dataset, as evidenced by cross-validation stability and AUC-ROC results.
4. A feature importance analysis identifying the most diagnostically relevant nuclear morphological features, which may inform future feature selection strategies in breast cancer classification research.

6.3 Future Work

Several promising directions exist for extending this research:

- **Hyperparameter Tuning:** Applying Grid Search or Bayesian Optimization to systematically optimize model parameters (e.g., the number of trees in Random Forest, the value of k in KNN, and the regularization strength C in Logistic Regression) may yield measurable performance improvements.
- **Advanced Algorithms:** Extending the comparison to include Support Vector Machines (SVM), Gradient Boosting (XGBoost, LightGBM), and deep learning architectures (e.g., feedforward neural networks) would provide a more comprehensive benchmarking of the algorithmic landscape for this classification task.
- **Dimensionality Reduction:** Applying Principal Component Analysis (PCA) or Recursive Feature Elimination (RFE) prior to classification — particularly for KNN — may improve model efficiency and address the multicollinearity identified among size-related features.
- **Class Imbalance Handling:** Incorporating oversampling techniques such as SMOTE or cost-sensitive learning to address the mild class imbalance may further improve Recall for the malignant class, which is the clinically critical minority class.
- **External Validation:** Evaluating the trained models on independent, external breast cancer datasets (e.g., SEER, NKI, or institutional clinical datasets) would provide stronger evidence of generalizability beyond the WDBC benchmark.
- **Explainability:** Incorporating model explainability frameworks such as SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations) would enhance the clinical interpretability of model predictions at the individual patient level, a prerequisite for responsible deployment in clinical decision-support systems.

6.4 Closing Remarks

Breast cancer continues to represent one of the most significant global public health challenges, with incidence and mortality rates projected to rise substantially by 2050 (Kim et al., 2025). The integration of machine learning into clinical diagnostic workflows offers a promising pathway toward earlier, more accurate, and more equitable detection of malignancies — particularly in resource-limited settings where access to specialist pathological expertise may be constrained. While the models developed in this study represent a foundational step, they demonstrate that even relatively simple and interpretable algorithms, when applied rigorously to well-characterized datasets, can achieve diagnostic performance approaching that of more complex methods. Continued research, clinical validation, and interdisciplinary collaboration between data scientists and healthcare professionals will be essential to translating these algorithmic advances into tangible improvements in patient outcomes.

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